

WON JUN LEE

Ph.D. Candidate in Electrical and Computer Engineering

PROFILE

I am a Ph.D. candidate in Electrical and Computer Engineering at Purdue University with expertise in **network scheduling**, stochastic optimization, and scalable algorithm design for communication-constrained systems, with a particular focus on **information freshness**. I am interested in research and engineering roles spanning **cloud and datacenter infrastructure**, **distributed systems**, **system optimization**, and related technical domains.

CONTACT DETAILS

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Website

ACADEMIC ACTIVITIES

Reviewer, IEEE International Symposium on Information Theory (ISIT), 2026

AWARDS & HONORS

Ross Fellowship, Purdue University, 2021–2025
Student Travel Grant, IEEE International Symposium on Information Theory (ISIT), 2024
Dean's List, Yonsei University, 2015–2016, 2019–2020

SKILLS

Programming Languages: Python (PyTorch), C/C++, MATLAB, Java/Android
Modeling & Optimization: Convex Optimization, Control, Information Theory, Stochastic Processes
Research Areas: Network Scheduling, Cloud Infrastructure, IoT, Information Freshness

EDUCATION

PURDUE UNIVERSITY

Ph.D. Candidate, Electrical and Computer Engineering
Advisor: Prof. Chih-Chun Wang

GPA: 3.81/4.0

West Lafayette, IN, U.S.
Aug 2021 – Present

YONSEI UNIVERSITY

B.S., Electrical and Electronics Engineering
GPA: 3.84/4.0

Seoul, South Korea
Mar 2015 – Feb 2021

RESEARCH EXPERIENCE

PROF. CHIH-CHUN WANG'S LAB

RESEARCH ASSISTANT, PURDUE UNIVERSITY

West Lafayette, IN, U.S.
Aug 2021 – Present

Latency-and-Cost-Aware Channel Scheduling → [C2]

- Developed a **scheduling and resource-allocation framework** for heterogeneous multi-channel systems, jointly optimizing latency, transmission cost, data quality, and **information freshness**.
- Designed a **structured channel-selection algorithm** that systematically computes the policy thresholds and enables scalable scheduling decisions in systems with diverse channel attributes.
- Reduced the computational complexity** of finding the scheduling policy from $O(N^3)$ to $O(N)$, improving the practicality of the approach for larger-scale systems.

Cost-Aware Sensor Scheduling in the Correlated Noise Environment

- Developed a **cost-aware scheduling framework** for multi-sensor systems under **correlated noise**, targeting efficient estimation under communication and sensing constraints.
- Designed an estimation method that **alleviates the computational burden** associated with **Kalman filtering** in correlated-noise settings, improving tractability for high-dimensional systems.
- Proposed an optimization approach for balancing estimation error and communication cost in **resource-constrained sensing systems**.

DIGITAL COMMUNICATIONS LAB

UNDERGRADUATE RESEARCH INTERN, YONSEI UNIVERSITY

Seoul, South Korea
Jan – Jun 2020

5G Network Performance Benchmark in Heterogeneous Environment

- Developed an Android-based network analysis application using **UDP socket programming** to measure end-to-end latency/throughput in heterogeneous network communication environments.
- Implemented **UDP hole punching** to support connectivity benchmarking in real-world 5G settings.

HIGH DIMENSIONAL SIGNAL PROCESSING LAB

UNDERGRADUATE RESEARCH INTERN, YONSEI UNIVERSITY

Seoul, South Korea
Jul – Aug 2019

DCN Classification Logic Analysis → [C1]

- Assisted in analyzing decision boundaries of deep neural network classifiers.

MILITARY EXPERIENCE

KATUSA (KOREAN AUGMENTATION TO THE UNITED STATES ARMY)

SERGEANT

Seoul, South Korea
Mar 2017 – Dec 2018

- Served as a de facto Information Management Officer (25B) at the USFK Public Affairs Office, supporting mission-critical IT and communications operations.
- Led 6 soldiers as a **squad leader** while managing approximately 100 network-connected workstations and coordinating technical operations between ROK and U.S. military teams during international joint military exercises (Ulchi-Freedom Guardian 2017 / Foal Eagle 2018).

PUBLICATION

- [C2]** W. J. Lee, C.-C. Wang, "Aol-optimal scheduling for arbitrary K-channel update-through-queue systems," Proc. IEEE Int. Symp. on Information Theory (ISIT), 2024, pp. 957–962
- [C1]** J. Kim, S. Woo, W. J. Lee, D. Kim, and C. Lee, "Analyzing Decision Polygons of DNN-based Classification Methods," Proc. 17th Int. Conf. Informatics in Control, Automation and Robotics (ICINCO), 2020, pp. 112–119